MathEXplainedMAGAZINE

New Year's Math Crossword Solutions

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ONE DOWN

To find the average area of a firework, we essentially need to calculate $10^{-1}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 5^2 + 4^2 + 3^2 + 2^2 + 1^2)\,\pi$. By noticing that the average area of the 6th - 10th fireworks is the same as the average area of the 1st - 5th fireworks, we can rewrite the expression as $5^{-1}(1^2 + 2^2 + 3^2 + 4^2 + 5^2)\,\pi = 11\pi$. Thus, our answer is 11 + 1 = **12**.

ONE ACROSS

Notice that f(x) is just a sine curve with a lot of transformations. Applying all of the transformations one by one, we can see that the section of f(x) from 0 to 31 is a full period with a min at x = 15.5 and maxes at x = 0, 31. Since f(x) is 50% at x = 15.5, we know that f(x) = 75% at the point halfway between 15.5 and 0 – 7.75 – and the point halfway between 31 and 15.5 – 23.25. Since there are 8 days in January larger than 23.25 and 7 days in January smaller than 7.75, our answer is just 8 + 7 = **15**.

THREE ACROSS

Let the roots of P(x) be r, s, and t. By Vieta's formulas, we have:

$r + s + t = -(2a + 5)$
$rs + rt + st = 25a$
$rst = -2025$

Since rst = -2025, and all roots are integers, it is logical to look for integer factors of -2025. The prime factorization of 2025 is $3^4 * 5^2$.

Thus, the possible integer roots are among the divisors of 2025. To minimize trial and error, we test manageable divisors first. Considering r = -5, we have:

$P(-5) = (-5)^3 + (2a+5)(-5)^2 + 25a(-5) + 2025 = -75a + 2025.$

Setting P (-5) = 0 since -5 is a root, we get that $-75a + 2025 = 0 \rightarrow a =$ **27**.

To verify, we substitute a = 27 back into the polynomial to get $P(x) = x^3 + 59x^2 + 675x + 2025$. We know that x = -5 is a root, so we can factor out (x+5) so that $P(x) = (x+5)(x^2 + 54x + 405)$. Factoring the quadratic equation, we arrive at $P(x) = (x+5)(x+9)(x+45)$. The roots are -5, -9, and -45, all of which are integers, as desired.

TWO DOWN

If there are n people at the party, then there is n C 2 ways to select a distinct combination of 2 people. Since each combination of 2 people shake hands twice, the total number of handshakes is 2 * n C 2. It then follows that 2 * n C 2 > 3100. Simplifying:

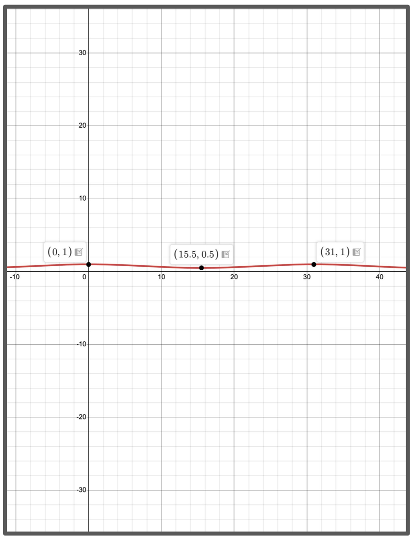
$n * (n-1) > 3100 \rightarrow n^2 - n - 3100 > 0.$

Applying the quadratic formula, we see that $n^2 - n - 3100$ reaches a value of 0 at approximately n = 56.18. Thus, our answer is $\lceil n \rceil =$ **57**.

AUTHOR'S NOTE

We hope you guys enjoyed these questions! Please let us know if anyone discrepancies in problem solutions exist by messaging in the #math-help channel of our Discord:
https://tinyurl.com/MEXplained

1. 1	2. 5
3. 2	7

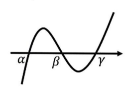


Vieta's Formulas

Cubic

Given α, β and γ are the roots of
$$ax^3 + bx^2 + cx + d = 0,$$

- Sum of roots : $\alpha + \beta + \gamma = -\frac{b}{a}$
- Sum of pairs : $\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$
- Product of roots: $\alpha\beta\gamma = -\frac{d}{a}$



- Sum of Squares: $\alpha^2 + \beta^2 + \gamma^2$
$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$
- Sum of Cubes : $\alpha^3 + \beta^3 + \gamma^3$
$= (\alpha + \beta + \gamma)^3 - 3(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \alpha\gamma) + 3\,\alpha\beta\gamma$